

*Teaching Notes 4B**

APV & WACC

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*Many thanks to Prof. Amit Seru who made his material available to me. These teaching notes closely follow his.

Overview

- Last time we discussed basic valuation
 - How to determine the FCF from a project
 - How to find r_A , the required return on the assets
 - We were in a world with NO effects of financing
- If we borrow (issue debt) to finance our project, we have to make interest payments
 - Interest is tax deductible: paid out of before-tax earnings
 - The resulting tax reduction (or Debt Tax Shield) is valuable
- Two extended approaches to valuation that account for the effects of financing
 - Weighted Average Cost of Capital (WACC)
 - Adjusted Present Value (APV)

Objective

- Understand the “APV” and “WACC” methods
- Understand the pros and cons of the two methods

Example to Cement Ideas: What is “DTS”?

Example: Tax Deductibility of Interest

- Firm U has no debt
- Firm L has borrowed \$1,000 at 8%

	Income Statement of Firm U	Income Statement of Firm L
Earnings before interest and taxes	\$1,000	\$1,000
Interest paid to bondholders	-	80
Pretax income	1,000	920
Tax at 35%	350	322
Net income to stockholders	650	598
Total income to both bondholders and stockholders	$\$0 + 650 = \650	$\$80 + 598 = \678
Interest tax shield ($.35 \times \text{interest}$)	\$0	\$28

Example (cont.): Tax Deductibility of Interest

- The value of a project is the value of the total cash flows to the investors (D+E)
 - Without leverage, the assets return \$650 to investors
 - With leverage, the same assets return \$678
- The \$28 represents an additional cash flow to investors when financing by borrowing
 - The value of the project arising from debt financing is called the debt tax shield (DTS)

General Comparison of WACC vs. APV

- Both methods involve discounting free cash flows
 - Both involve the Balance Sheet
- WACC method adjusts the discount rate to reflect the effects of financing (debt tax shield) on value
 - Uses free cash flows as before
 - Does not use r_A as discount rate; instead, uses weighted average cost of capital (WACC) as discount rate
- APV method adjusts the cash flows to reflect the effects of financing (debt tax shield) on value
 - Uses r_A to discount cash flows from assets, as before
 - Adds cash flows to reflect the debt tax shield; uses either r_A or r_D to discount cash flows from debt tax shield

WACC: Weighted Average Cost of Capital

Weighted Average Cost of Capital (WACC)

- A single rate to discount the free cash flows and account for the tax benefit of debt
- Intuition behind WACC
 - A project (or a firm) must generate sufficient returns to compensate investors (equity holders and debt holders) for risk
 - Interest payments reduce taxes and hence reduce the required rate of return on the assets (to a rate below r_A)

WACC Formula

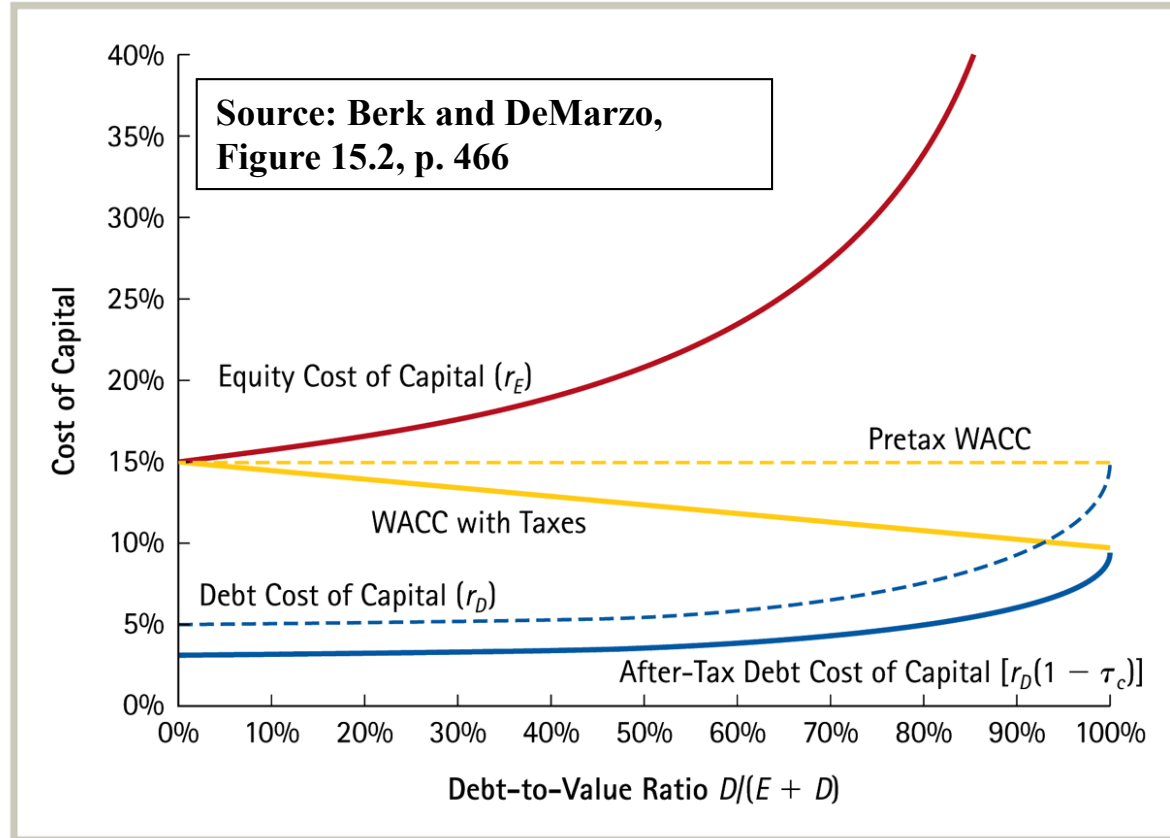
- The weighted-average cost of capital is

$$WACC = r_E \frac{E}{E + D} + r_D(1 - t_c) \frac{D}{E + D}$$

where t_c is the corporate tax rate.

- Intuitively, tax deductibility of interest means that ASSETS only need to earn $r_D(1-t_c)$ to pay debt holders a return of r_D

WACC Decreases as D/E Increases



- $D/(D+E)$ that the project will have going forward
 - What if the $D/D+E$ changes over time?

How to calculate WACC if D/E changes?

- When valuing projects, D/E , r_D and r_E should reflect future risk and leverage
- If D/E changes, then need to recalculate WACC at each point when the change happens
- Rough guidelines to follow:
 - Estimate r_D at new D/E (often we use old r_D)
 - To get new r_E , first determine r_A (often by unlevering an appropriate r_E)
 - Relever r_A using future D/E to get new r_E
- Use new D/E , r_D and r_E to calculate new WACC

APV: Adjusted Present Value

Adjusted Present Value (APV)

- Three Steps for Calculating APV
 1. Do valuation assuming 100% equity finance
 2. Determine additional discounted cash flows from the debt tax shields (including terminal value)
 3. Account for other financing effects (if any), such as costs of financial distress or loan subsidies

$$APV = NPV_{\text{all equity}} + PV(\text{financing choices})$$

The Value of the Debt Tax Shield

- Interest payment is tax deductible. Reduce tax payments by:
 $t_c \times \text{interest}$
 - Additional cash flow period by period
 - Discount at an appropriate rate
- Three candidates for discount rates:
 1. r_D , since this reflects the risk of the debt
 2. r_A , since this reflects the risk of the assets generating profits
 3. r_E , since it is conservative
- Depends on debt policy of the firm
 - If debt is not too high and is fixed, r_D seems appropriate
 - If D:E is constant, then r_A seems accurate
- If using r_D with a perpetual constant D, there is an easy formula: $PV[DTS] = t_c D$

Other Financing Effects

- Usually stop with the tax benefit of debt
- Other potential financing effects
 - Easier to quantify: loan guarantees, subsidized loans, costs of issuing securities, real options, etc.
 - Harder to quantify: costs of financial distress, agency costs, etc.

Some “Subtle” Points

Important Note about Terminal Values

- Must account for the terminal value of tax shield using the perpetuity method
 - For WACC: just use the WACC rate in the growing perpetuity formula
 - For APV: can calculate separate terminal value for cash flows from tax shield

Pros of WACC

- The inputs are easy to approximate
 - Particularly straightforward when the firm is targeting a fixed capital structure after the project's initiation
- When D/E and r_E are constant, WACC calculation is fairly accurate

Cons of WACC

- Difficult to accurately adjust for changes in risk or leverage
 - Easier to work with WACC when the project supports a constant future D/E
 - Assumes all benefits of tax shield accrue every period
- Not as flexible or transparent as APV

Pros of APV

- More transparent about sources of value
 - Separates the value of the assets from the value created by the financing decision
- More flexible
 - APV can use different discount rates for different components with different risks
 - In contrast to WACC, can account for other financing effects besides tax shield
 - Frequently employed in M&A, LBOs, etc.

Cons of APV

- Requires explicitly deriving schedule of future debt levels and interest payments

An Example: WACC & APV Applied

Example: Sangria's Use of WACC

- Sangria promotes a happy, low-stress lifestyle
 - Cost of debt (r_D) = .06
 - Cost of equity (r_E) = .124
 - Marginal tax rate (t_C) = .35
 - Debt ratio ($D/D+E$) = $500/1,250 = .4$
 - Equity ratio ($E/D+E$) = $750/1,250 = .6$
- $WACC = .124 \times .6 + .06 \times (1-.35) \times .4 = 9\%$

Sangria's Use of WACC (cont.)

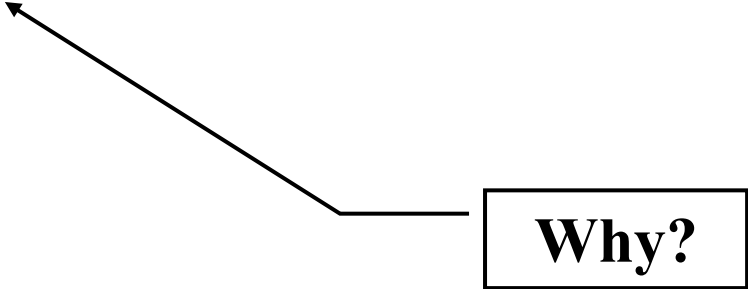
- Sangria considers investing \$12.5m in a perpetual crushing machine
 - CF of \$1.731m before tax for perpetuity
 - FCF = \$1.125m after tax
- Project has same risk as Sangria's other operations and it is financed with the same debt and equity ratios (E = \$7.5m and D = \$5m)
- $NPV = -\$12.5m + \$1.125m / 9\% = 0$
 - Project just exactly breaks even

Example: APV of Sangria's Crusher

- Remember, we consider investing \$12.5m in crusher, generating after tax cash flow of \$1.125m
- We know: $r_D = 6\%$, $r_E = 12.4\%$, $D/D+E = 40\%$, and $t_c = 35\%$
- WACC calculation found $NPV = 0$
- What is APV?

APV of Sangria's Crusher (cont.)

- First, calculate NPV as if investment is all equity financed
 - $r_A = 12.4\% \times 60\% + 6\% \times 40\% = 9.84\%$
 - $NPV_{ALL\ E} = -\$12.5m + \$1.125m/9.84\% = -\$1.067m$
- Value of Debt Tax Shield
 - Each period, interest payment is 6% of \$5m = \$.3m
 - Tax reduction is \$.3m x 35% = \$.105m
 - $PV(DTS) = \$.105m / 9.84\% = \$1.067m$
- $APV = -\$1.067m + \$1.067m = 0$



Why?

Extensions of Unlevering/Magical Formula

Unlevering with Taxes

- Earlier, we saw the equation for unlevering equity betas to get asset betas
- With taxes, the firm's debt policy affects the unlevering formula (two cases)
 - Debt constant or
 - D/E constant

Levering/Unlevering w/ Taxes:

D/E fixed

$$\beta_A = \beta_E \left(\frac{E}{E + D} \right) + \beta_D \left(\frac{D}{E + D} \right)$$

When the company you are levering/unlevering maintains a fixed debt to total capital ratio in market values – in presence of corporate taxes

Note: same as TN 4B with no taxes!

Levering/Unlevering w/ Taxes:

D fixed

$$\beta_A = \beta_E \left(\frac{E}{E + (1 - t_c)D} \right) + \beta_D \left(\frac{(1 - t_c)D}{E + (1 - t_c)D} \right)$$

$$\Rightarrow \beta_E = \beta_A + \frac{D}{E} (\beta_A - \beta_D)(1 - t_c)$$

When the company you are levering/unlevering maintains a fixed determined amount of debt – in presence of corporate taxes

Takeaways: APV and WACC

- Interest payments are tax deductible and the resulting debt tax shield is valuable
- APV accounts for this by adjusting cash flows
 - Very flexible tool
- WACC accounts for this by adjusting discount rate
 - Less flexible but faster
- With taxes, two methods for unlevering/levering, depending on the firm's debt policy